Junyoung Park

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RESEARCH INTERESTS

- Inverse problems and parameter estimation in structural systems
- Probabilistic modeling of uncertainty in structural measurements and models
- Structural condition assessment from continuous field measurements
- Long-term field monitoring of in-service structures (bridges, pavements, ports, precast members)

EDUCATION

Chung-Ang University **2022 – 2024**
M.S. in Civil Engineering (Structural Engineering) · *Seoul, Republic of Korea*
- Thesis: A Scalable Bridge Health Monitoring System using an IoT Sensor and Cloud Computing
- Advisor: the late Prof. Jong-Woong Park · GPA 4.39 / 4.5

Chung-Ang University **2017 – 2022**
B.E. in Civil Engineering & B.S. in Mathematics (Double Degree) · *Seoul, Republic of Korea*
- Thesis: Development of a Low-power IoT Sensor and a Cloud-based Data Fusion Displacement Estimation Method for Ambient Bridge Monitoring
- GPA 4.20 / 4.5 (summa cum laude)
- Relevant coursework: Analysis I–II, Partial Differential Equations, Differential Equations I–II, Numerical Analysis, Probability and Statistics, Differential Geometry I–II

RESEARCH EXPERIENCE

Researcher | UDNS Co., Ltd., R&D Division **2024 – Present**
Industrial Technical Personnel Program (R&D-based alternative to military service) · *Republic of Korea*

Plate-type high-speed weigh-in-motion (HS-WIM) system development and field validation (MOLIT) 2024 – Present
- Designed the multi-channel signal acquisition PCB for the load cells
- Developed signal-processing algorithms that extract axle load, gross weight, speed, and vehicle dimensions from raw sensor signals, combining automatic load-segment detection with FIR-filtered dynamic weight estimation
- Conducted on-road field tests with loaded trucks to reduce estimation error across vehicle class, speed, and payload

HS-WIM signal analysis and accuracy improvement (MOLIT) 2024 – Present
- Developed deep-learning-based fault-tolerant reconstruction of multi-channel HS-WIM signals
- Analyzed paired HS-WIM and on-board mass signals toward auto-calibration without a reference scale
- Analyzed traffic patterns from a year of continuous HS-WIM measurements, characterizing vehicle class and gross weight by time of day and season

On-board mass (OBM) system for real-time heavy-vehicle load monitoring 2024 – Present
- Improved dynamic weight estimation accuracy by estimating payload from multi-channel suspension sensors and compensating vehicle dynamic effects with IMU signals
- Conducted environmental, EMC, and ingress-protection testing for TCA Category B type approval
- Defined the communication protocol between the network module and the mobile application, and modified the Android application accordingly

Graduate Researcher | Smart Infrastructure Technology Laboratory **2019 – 2024**
Chung-Ang University · *Seoul, Republic of Korea*

Long-term monitoring of in-service bridges (Seoul Metropolitan Government; M.S. thesis) 2020 – 2024
- Designed JANET, a low-power IoT sensor: 24-bit three-axis accelerometer, eight strain channels, 0.34 mA standby, direct LTE Cat-1 communication
- Instrumented Seongsan, Paldang, Seosomun, and Jamsu bridges and collected over one year of continuous data under live traffic at 1 kHz across eight channels (3-axis acceleration, four strain, temperature)
- Built and operated a scalable data pipeline and dashboards serving all four bridges from a single cloud back end
- Tracked natural frequencies over a year, where seasonal temperature variation dominated the observed shifts
- Validated sensor-fusion displacement estimation against a laser vibrometer reference, achieving under 1% error

Bridge load testing and load-carrying capacity evaluation (MOLIT) 2020 – 2024
- Developed a portable multimetric sensing system (JANET) and web-based document automation for capacity evaluation
- Devised an automatic load-event interval recognition method for continuous load-carrying capacity assessment

Monitoring of precast members during transportation and construction (MOLIT) 2020 – 2024
- Developed a portable gateway-and-JANET system for real-time safety monitoring of prefabricated members in transit
- Monitored acceleration, tilt, strain, and temperature throughout lifting, loading, and transport, with chatbot anomaly alerts

Strain visualization module with E-paper QR display (MOLIT) 2021 – 2024
- Developed an ultra-low-power multi-channel module that records strain history from fabrication to service and holds it on an embedded E-paper display that remains readable with no power

– Built a chatbot-based QR acquisition pipeline and automated reporting

Smart port berthing monitoring (MOLIT) 2021 – 2023
– Developed a wireless sensing system for long-term monitoring of caisson-type quay walls, integrating a high-sensitivity accelerometer with tilt, temperature, camera, and rangefinder modules
– Triggered vibration measurement on vessel berthing and unberthing, with an AI classifier to identify quay-wall events

Self-stress sensing circuit for ultra-high-performance concrete (NRF) 2020 – 2021
– Designed a low-cost multi-channel wireless sensing circuit that infers the loading state of self-sensing concrete from resistance change

Low-cost IoT seismic sensor (NRF) 2019
– Designed an ultra-low-power seismic sensor built for under $50 per unit
– Developed a two-phase detection algorithm: STA/LTA screening in the time domain, then classification by spectral content

Visiting Researcher | University of Hawaii at Manoa 2022
Department of Civil, Environmental and Construction Engineering · Honolulu, HI, USA · two-week research visit
– Implemented GPS-based time synchronization on JANET for multi-point road-surface vibration measurement
– Configured and demonstrated the measurement setup for the host lab, and gave a lab seminar on the sensing platform

JOURNAL PUBLICATIONS

Peer-reviewed
– **Park, J.**, Kim, J. W., Cho, Y., & Jung, Y. (2026). YOLO-Based Automatic Load Segment Detection for High-Speed Weigh-in-Motion Systems. *Journal of the Korean Society of Civil Engineers*, 46(1), 95–104. (in Korean)
– Shin, J., **Park, J.**, Won, J., Park, J. W., & Min, J. (2024). Development of AI-Based Multisensory System for Monitoring Quay Wall Events. *Journal of Marine Science and Engineering*, 12(11), 1902.
– **Park, J.**, Kim, J. S., Kim, J. W., & Cho, Y. (2024). Stabilization of Dynamic Weight Estimation Systems for Commercial Vehicles during Driving. *Journal of the Korean Society of Mechanical Technology*, 26(5), 1041–1046. (in Korean)
– Khayam, S. U., Ajmal, A., **Park, J.**, Kim, I. H., & Park, J. W. (2023). Tendon Stress Estimation from Strain Data of a Bridge Girder Using Machine Learning-Based Surrogate Model. *Sensors*, 23(11), 5040.
– Khayam, S. U., Won, J., Shin, J., **Park, J.**, & Park, J. W. (2023). Monitoring Precast Structures during Transportation Using a Portable Sensing System. *Automation in Construction*, 145, 104639.
– Won, J., Park, J. W., **Park, J.**, Shin, J., & Park, M. (2021). Development of a Reference-Free Indirect Bridge Displacement Sensing System. *Sensors*, 21(16), 5647.
– Le, H. V., Kim, T. U., Khan, S., **Park, J.**, Park, J. W., Kim, S. E., … & Kim, D. J. (2021). Development of Low-Cost Wireless Sensing System for Smart Ultra-High Performance Concrete. *Sensors*, 21(19), 6386.
– Khan, S., Won, J., Shin, J., **Park, J.**, Park, J. W., Kim, S. E., … & Kim, D. J. (2021). SSVM: An Ultra-Low-Power Strain Sensing and Visualization Module for Long-Term Structural Health Monitoring. *Sensors*, 21(6), 2211.
– **Park, J.**, Shin, J., Won, J., Park, J. W., & Park, M. (2021). Development of a Low-Power IoT Sensor and Cloud-Based Data Fusion Displacement Estimation Method for Ambient Bridge Monitoring. *Journal of the Computational Structural Engineering Institute of Korea*, 34(5). (in Korean)
– Won, J., **Park, J.**, Park, J. W., & Kim, I. H. (2020). BLESeis: Low-Cost IoT Sensor for Smart Earthquake Detection and Notification. *Sensors*, 20(10), 2963.

Under review
– **Park, J.**, Shin, J., & Won, J. Development of a Portable Multimetric Sensing System for Bridge Load Testing and Load-Carrying Capacity Evaluation. *Measurement* (under review), 2026.
– **Park, J.**, Jho, Y., & Kim, J. W. Comparative Performance of Cloud and Local Models in Hybrid LLM Question Answering for WIM Operational Data in Freight Transport Safety. *Journal of the Korean Society of Civil Engineers* (under review), 2026. (in Korean)
– Kim, Y., Jung, H., **Park, J.**, Lee, Y., Jung, Y., Kim, J. W., & Yoon, I. Analysis of Freight Vehicle Operation Restriction Violations and Improvement of the Enforcement System Based on Real-World Road Data. *The Journal of the Korea Institute of Intelligent Transport Systems* (under review), 2026. (in Korean)
– Khan, S., **Park, J.**, Park, J. W., Kim, I. H., & Won, J. Low-Power Multi-Channel Strain Sensing and Instant Shape Visualization for Digital Twin. *Journal of Civil Structural Health Monitoring* (under review), 2026.

In preparation
– **Park, J.**, Jho, Y., & Kim, J. W. Fault-Tolerant Reconstruction of Multi-Channel Weigh-in-Motion Signals Using Deep Learning. Target: *IEEE Transactions on Instrumentation and Measurement*.

CONFERENCE PRESENTATIONS (FIRST AUTHOR)

International
– **Park, J.**, Shin, J., & Park, J. W. (2023). Cloud-Based Ambient Bridge Monitoring using Strain and Acceleration Measurements. *ITC*, Korea.
– **Park, J.**, Shin, J., Jang, H., & Park, J. W. (2022). Development of an IoT Sensor System and Web-Based Document Automation for Bridge Load-Bearing Capacity Evaluation. *CONVR*, Korea.

Domestic
– **Park, J.**, Kim, J. H., & Kim, J. W. (2025). Development of a Cloud-Based Integrated Platform for Autonomous Enforcement. *KSCE Annual Conference*.
– **Park, J.**, Shin, J., & Park, J. W. (2023). A Scalable Framework for Ambient Bridge Monitoring using IoT Sensor and Cloud Computing. *KSMI Annual Conference*. **— Best Paper Presentation Award**

- **Park, J.**, & Park, J. W. (2023). IoT Sensor and Cloud Computing based Long-Term Bridge Monitoring. *KIBSE Annual Conference*.
- **Park, J.**, Park, J. W., & Shin, J. (2023). Bridge Monitoring System using an IoT Sensor and a Cloud Server. *KSNVE Annual Conference*.
- **Park, J.**, Shin, J., Park, J. W., & Park, M. (2022). Optimization of an Acceleration–Strain Displacement Estimation Method Considering Bridge Boundary Conditions. *COSEIK Annual Conference*. (also presented at KSNVE Annual Conference, 2022)
- **Park, J.**, Shin, J., Kang, S., Won, J., & Park, J. W. (2021). Cloud-Based Long-Term Bridge Monitoring System. *KSMI Annual Conference*.
- **Park, J.**, Shin, J., Won, J., Park, J. W., & Jang, Y. (2021). Estimation of a Bridge's Damaged Cable Using Deep Learning. *KSMI Annual Conference*.
- **Park, J.**, Shin, J., Won, J., Park, J. W., & Park, M. (2021). Cloud-Based Bridge Displacement Monitoring System. *COSEIK Annual Conference*.

Accepted

- **Park, J.**, Song, M., Jho, Y., & Kim, J. W. Cross-Analysis of WIM and OBM Signals for Dynamic Load Assessment and Measurement Error Correction. *10th International Conference on Weigh-In-Motion (ICWIM-10) / 19th International Symposium on Heavy Vehicle Transport Technology (HVTT-19)*, May 2027.

PATENTS

- Kim, J. W., Jho, Y., Park, J., & Song, M. Method and Apparatus for Estimating Combined Container Weight Information Using an On-Board Mass System and a Weigh Station. In preparation.
- Kim, J. W., Jho, Y., Jung, Y., & Park, J. Method and Apparatus for Vehicle Recognition Using a LiDAR Sensor. KR 10-2026-0209028, filed 2026.
- Kim, J. W., Jho, Y., Jung, Y., & Park, J. Vehicle Specification Measurement System and Method. KR 10-2026-0101228, filed 2026.
- Kim, J. W., Jho, Y., & Park, J. System and Method for Continuous Evaluation of Bridge Load-Carrying Capacity. KR 10-2025-0174354, filed 2025.
- Kim, J. W., Jho, Y., & Park, J. Method and Apparatus of Real-Time Compensation of Commercial Vehicle's Driving Mass Based on an FIR Filter. KR 10-2951213, registered 2026.
- Park, J. W., Park, J., & Shin, J. Automatic Load-Event Interval Recognition Using IoT Sensors for Structural Capacity Assessment. KR 10-2853678, registered 2025.
- Park, J. W., Kim, K. H., & Park, J. Concrete Crack Trajectory Monitoring and Long-Term History Visualization. KR 10-2892584, registered 2025.
- Park, J. W., Won, J., Park, J., & Shin, J. Cloud-Based IoT Sensor Data Preprocessing and Deep-Learning Anomaly Detection. KR 10-2705761, registered 2024.
- Park, J. W., Park, J., Shin, J., & Lee, J. IoT Sensor-Based Integrated Monitoring for Precast Transportation. KR 10-2558124, registered 2023.
- Park, S., Shin, J., & Park, J. Citizen-Participatory Smart City System Using Sensor Measurement and Data Visualization. KR 10-2389222, registered 2022.
- Park, J. W., & Park, J. Smart IoT Earthquake Sensor Unit and Notification Method. KR 10-2353297, registered 2022.

AWARDS

- **Best Paper Presentation Award**, Korea Institute for Structural Maintenance and Inspection **2024**
- **Gold Award**, X-Corps Plus Festival, NRF **2022**
- **Innovation Award**, Smart Construction Challenge (IoT / AI / Sensing), MOLIT · Korea National Railway **2021**
- **Minister of the Interior and Safety Award** — Smart Earthquake Detection Competition **2019**

Scholarships: CAU Graduate Research Scholarship (full graduate tuition, 2022–2023); SeAH Haiam Scholarship (full undergraduate tuition, 2020–2021); CAU Academic Excellence Scholarships (Fall 2017, Spring 2018).

MENTORING

- Mentored two undergraduate research projects, Undergraduate Research Program (URP), Chung-Ang University
- *Long-term monitoring of quay walls with wireless sensors* (Junsik Shin, 2023) — high-sensitivity accelerometers with AI-based vessel detection; later developed into a peer-reviewed paper
- *Vision-based long-term 6-DOF monitoring of bridge bearings* (Gunhee Kim, 2022)

SKILLS & LANGUAGES

Programming: Python, MATLAB, C, JavaScript / Node.js
Machine learning: PyTorch, scikit-learn, pandas, NumPy, YOLO
Hardware & design: Altium Designer, SolidWorks, Simulink, LabVIEW
Software & data: Android and web development, AWS, Docker, MySQL, InfluxDB, Grafana
Languages: English (TOEFL iBT 100/120, C1, 2026), Korean (native)

REFERENCES

Changsu Shim, Ph.D.

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